Lead-Lag Relationships Between Regional Oil Prices, US Retail Oil Prices, and Crude Oil Prices

Ellie Huang Nikita Jayaprakash Annie Wang Aishani Pradhan
Steve Wong

Executive summary

...

1 Introduction

This analysis explores gasoline price dynamics by analyzing the relationship of American gasoline prices with crude oil prices and comparing regional gasoline prices within the United States. The main research questions are:

- How do Brent crude oil price changes pass through to U.S. retail gasoline prices, and over what time range does this occur?
- How do regional gasoline prices co-move over time? Are there lead–lag relationships between regions?

Based on the findings, changes in the price of Brent crude oil affect the price of gasoline in the US at a one-week lag. Regional gasoline prices are correlated and move in conjunction with each other, especially in the Midwest, Gulf Coast and East Coast. Gasoline prices in the West Coast and the Rockies regions have less correlation with the other regions, especially during larger shocks,implying more local impacts to the price. The West Coast and Rockies regions may have higher transportation-related costs and policy-driven costs, such as tax. Forecasts predict a decrease in gasoline and Brent crude oil prices in the short term with the best prediction being the mean in the long term.

1.1 Data

ADD CITATIONS

The main data sources are the weekly U.S. gasoline prices from the U.S. Energy Information Administration and Brent crude oil prices from Yahoo Finance. The weekly U.S. gasoline prices include gasoline prices for overall country and regional prices. The time period for the data is from 1994 to 2025. Brent crude is the main input cost for gasoline, so changes in oil prices often pass through to pump prices with a lag. Regional gasoline data also captures differences in refining capacity, transportation, and local supply constraints. The regions are East Coast, Midwest, Gulf Coast, Rocky Mountains, and West Coast. This analysis helps explain how crude oil shocks affect consumer fuel prices and inflation. Forecasting this relationship can improve pricing expectations, support policy and business planning, and help anticipate the impact of supply disruptions or geopolitical events.

2 Methods

Text example that references Table 1.

Table 1: Passenger gender and class divisions on the Titanic. More 3rd class passengers were males seeking a better life.

Passenger class	female	male
1	94	122
2	76	108
3	144	347

3 Results

Text example that references Figure 1.

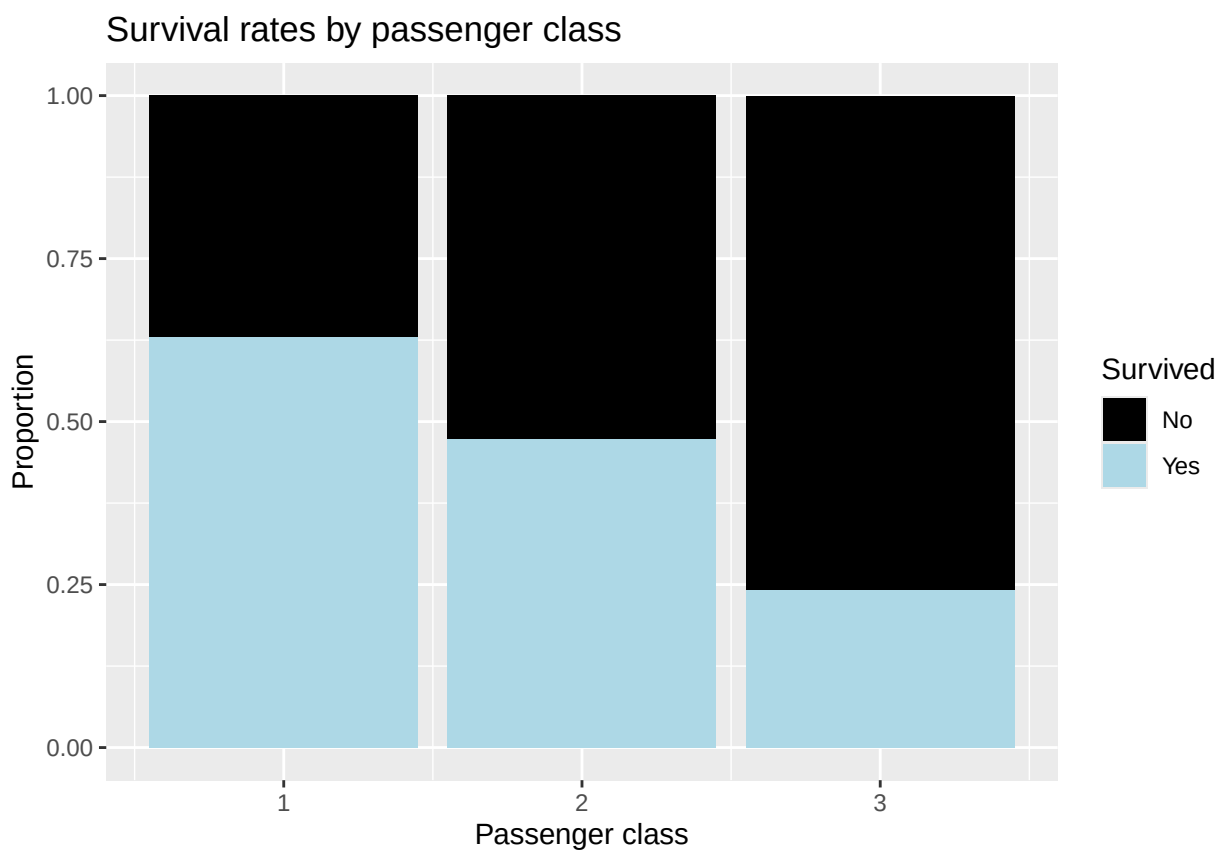

Figure 1: Survival rates across different passenger classes. This figure does not tell the full story as Table 1 shows that more of the lower class passengers were male.

4 Discussion

...

Appendix

Notice how the appendix below gathers all the code blocks above and nicely pastes them together.

```r
###########################
# STYLE EDITS: IGNORE THIS
###########################

# normally you'll want to include this with the libraries at the beginning of your document
knitr::opts_chunk$set(message = FALSE) # include this if you don't want markdown to knit messages
knitr::opts_chunk$set(warning = FALSE) # include this if you don't want markdown to knit warnings
knitr::opts_chunk$set(echo = FALSE) # set echo = FALSE to hide code from output

library(pander)
library(tidyverse)
library(reshape2)
library(kableExtra)
library(knitr)

# Titanic table

titantic <- read_csv(paste0("https://raw.githubusercontent.com/",
                            "benjaminleroy/stat315summer_data/",
                            "master/assignments/assignment03/titanic.csv"))
titantic %>% group_by(Pclass, Sex) %>%
  summarize(t = n()) %>% dcast(Pclass ~ Sex) %>%
  rename(`Passenger class` = Pclass) %>%
  kable(format = "latex",
        caption = paste("Passenger gender and class divisions on the Titanic.",
                        "More 3rd class passengers were males seeking a",
                        "better life."),
        booktabs = T) %>%
  kable_styling(latex_options = "HOLD_position")

# Titanic visual

titantic %>% ggplot() +
  geom_bar(aes(fill = factor(Survived,
                             levels = c(0,1),
                             labels = c("No", "Yes")), Pclass),
           position = "fill") +
  scale_fill_manual(values = c("black", "lightblue")) +
  labs(fill = "Survived",
       y = "Proportion",
       x = "Passenger class",
       title = "Survival rates by passenger class")
```